

Stats 101 Assignment

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1 DATASET

1.1 Dataset Selected

Mental Health in Tech Survey: Survey on Mental Health in the Tech Workplace in 2019

2 DATASET EXPLORATION

2.1 Data Cleanup

- I kept only the three dependent variables and the protected-class variables used as independent variables, and removed all other columns that were not needed for analysis.
- I removed 2 duplicate entries from the dataset.
- The column ‘What is your gender?’ had many inconsistent labels for the same gender (e.g., variations of male, female, and non-binary), which produced messy charts. I standardized these values into three categories — Male, Female, and Non-binary — in a ‘gender_clean’ column to make the charts clearer and easier to read.

2.2 Number of Observations

302

2.3 Number of Variables

9

2.4 Regulated Domain in Law

This dataset belongs to the employment/workplace domain. It collects information about mental health treatment, workplace disclosure, supervisor comfort, and industry support for all employees. Because the data includes variables tied to legally protected classes (age, gender, race, national origin, disability), it is subject to laws such as:

- Age Discrimination in Employment Act (ADEA), 1967 — age

- Civil Rights Act of 1964, Title VII — race, sex/gender, national origin
- Americans with Disabilities Act (ADA), 1990 — mental health as disability

2.5 Protected Class Table

Variable	Protected Class	Law
What is your age?	Age	Age Discrimination in Employment Act (ADEA), 1967
What is your gender?	Sex / Gender	Civil Rights Act of 1964, Title VII
What is your race?	Race	Civil Rights Act of 1964, Title VII
What country do you live in?	National Origin	Civil Rights Act of 1964, Title VII
Do you currently have a mental health disorder?	Disability	Americans with Disabilities Act (ADA), 1990
Do you have a family history of mental illness?	Genetic / Family Medical History	Genetic Information Nondiscrimination Act (GINA), 2008

3 RELATIONSHIPS BETWEEN DEPENDENT AND INDEPENDENT VARIABLES

3.1 Have you ever sought treatment for a mental health disorder from a mental health professional? x What is your age?

Group	False	True
< 30	38	53
30-39	45	80
40-49	22	46
50+	6	11

Table 1—Treatment and age

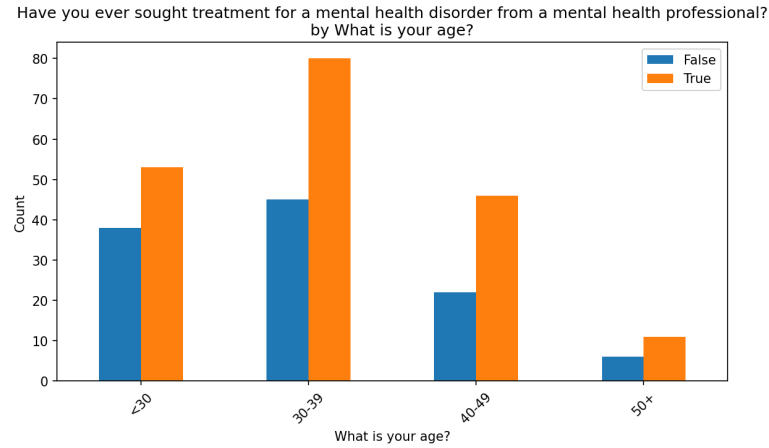


Figure 1—Treatment and age

3.2 Have you ever sought treatment for a mental health disorder from a mental health professional? x What is your gender?

Gender	False	True
Female	22	62
Male	82	117
Non-binary	2	9

Table 2—Treatment and gender

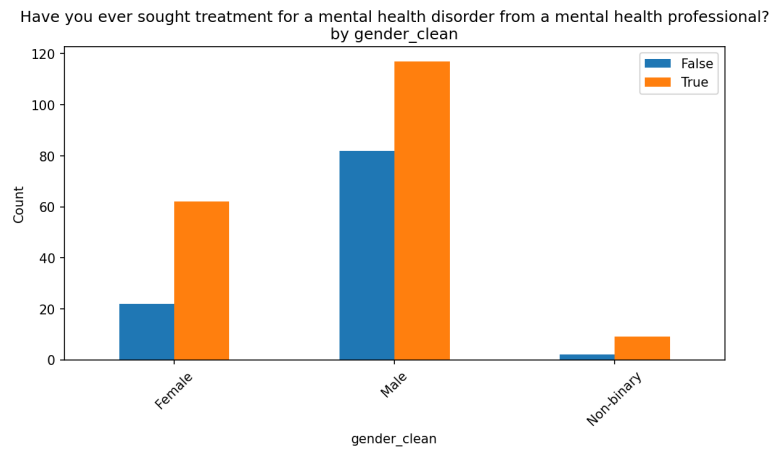


Figure 2—Treatment and gender

3.3 Have you ever sought treatment for a mental health disorder from a mental health professional? x What is your race?

What is your race?	False	True
Asian	6	3
Black or African American	0	2
Caucasian	0	1
Hispanic	1	0
I prefer not to answer	1	4
More than one of the above	2	6
White	40	116
White Hispanic	1	0

Table 3—Treatment and race

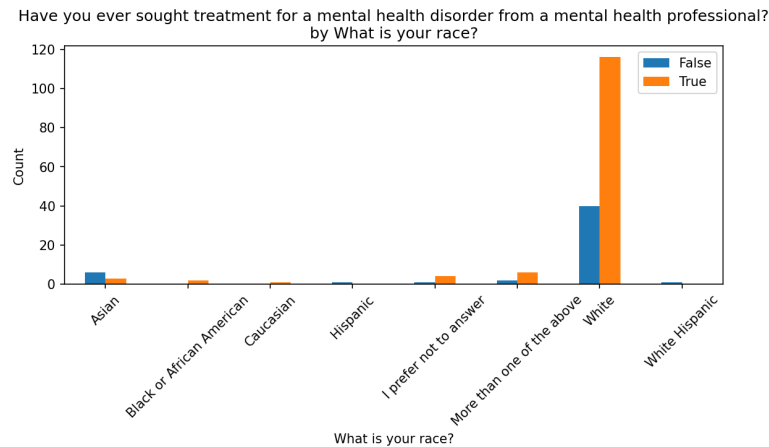


Figure 3—Treatment and race

3.4 Would you feel comfortable discussing a mental health issue with your direct supervisor(s)? x What is your age?

Group	Maybe	No	Yes
< 30	23	36	32
30–39	41	33	51
40–49	21	19	28
50+	9	4	4

Table 4—Disclosure and age

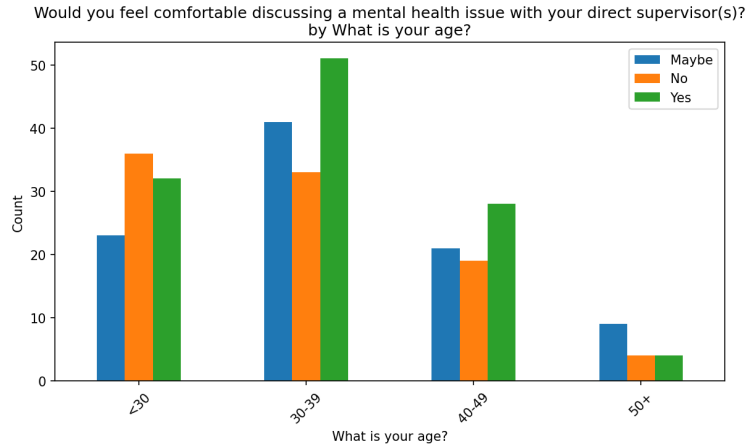


Figure 4—Disclosure and age

3.5 Would you feel comfortable discussing a mental health issue with your direct supervisor(s)? x What is your gender?

Gender	Maybe	No	Yes
Female	24	35	25
Male	64	52	83
Non-binary	4	3	4

Table 5—Disclosure and gender

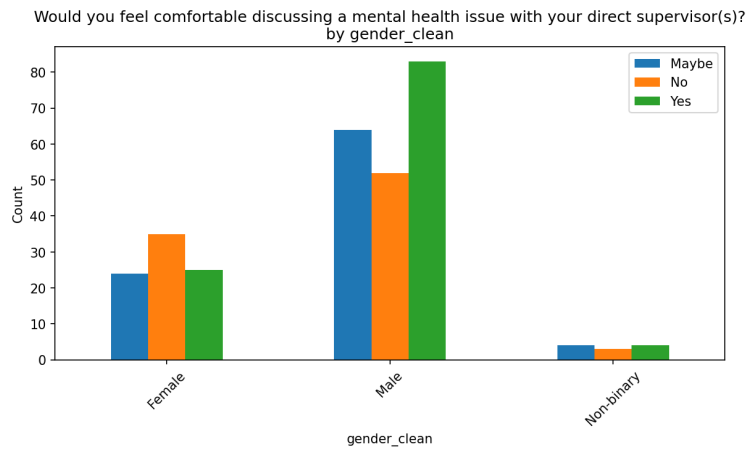


Figure 5—Disclosure and gender

3.6 Would you feel comfortable discussing a mental health issue with your direct supervisor(s)? x What is your race?

What is your race?	Maybe	No	Yes
Asian	4	3	2
Black or African American	2	0	0
Caucasian	0	1	0
Hispanic	0	1	0
I prefer not to answer	3	0	2
More than one of the above	1	6	1
White	47	40	69
White Hispanic	1	0	0

Table 6—Disclosure and race

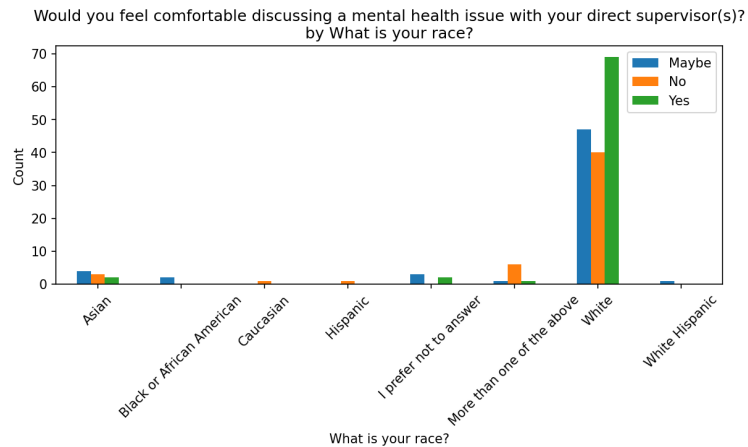


Figure 6—Disclosure and race

3.7 Overall, how well do you think the tech industry supports employees with mental health issues? x What is your age?

Group	1	2	3	4	5
< 30	9	32	35	13	2
30–39	14	37	56	15	3
40–49	16	18	23	11	0
50+	4	7	5	1	0

Table 7—Support and age

Overall, how well do you think the tech industry supports employees with mental health issues?
by What is your age?

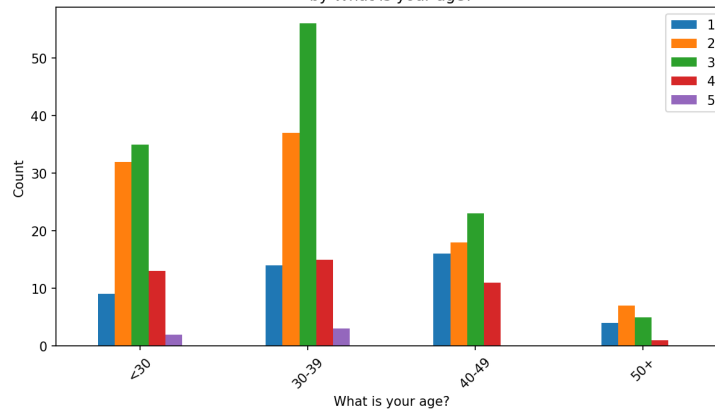


Figure 7—Support and age

3.8 Overall, how well do you think the tech industry supports employees with mental health issues? x What is your gender?

Gender	1	2	3	4	5
Female	12	22	41	8	1
Male	29	64	70	33	3
Non-binary	2	3	6	0	0

Table 8—Support and gender

Overall, how well do you think the tech industry supports employees with mental health issues?
by gender_clean

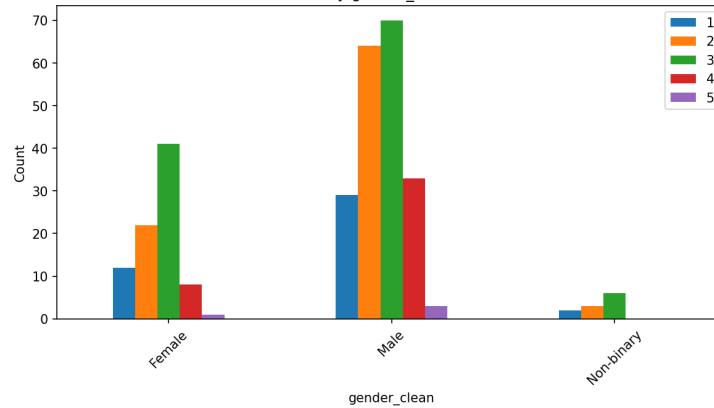


Figure 8—Support and gender

3.9 Overall, how well do you think the tech industry supports employees with mental health issues? x What is your race?

What is your race?	1	2	3	4	5
Asian	2	1	3	3	0
Black or African American	1	0	1	0	0
Caucasian	0	0	1	0	0
Hispanic	0	1	0	0	0
I prefer not to answer	1	2	1	1	0
More than one of the above	3	2	2	1	0
White	19	43	68	24	2
White Hispanic	0	0	1	0	0

Table 9—Support and race

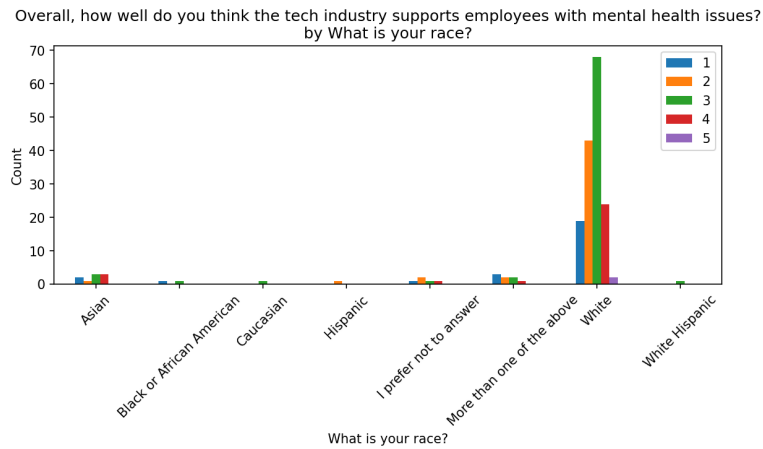


Figure 9—Support and race

4 MANIPULATION OF DATA

4.1 Fair Hypothesis

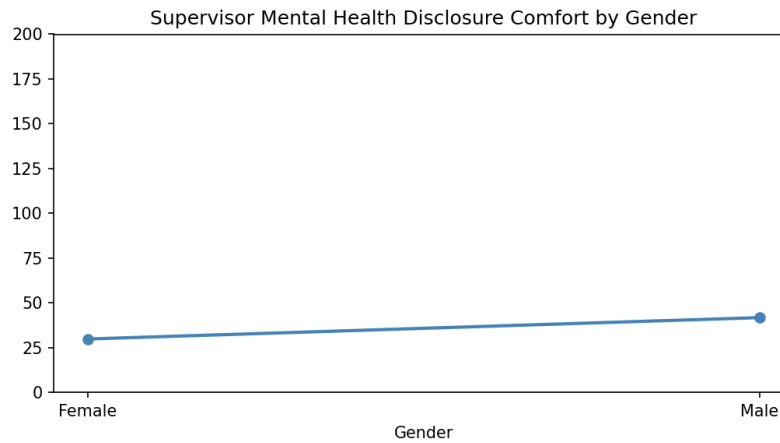


Figure 10—Fair disclosure

Fair hypothesis: As shown in the line chart, comfort discussing mental health with supervisors appears similar across genders, suggesting the workplace is fair.

Manipulations: I used a line chart and plotted only the “Yes” responses. I set the y-axis from 0 to 200 so both points sit low on the chart and the line looks almost flat. I also left off the y-axis label so it is less obvious that the values are percentages.

4.2 Bias Hypothesis

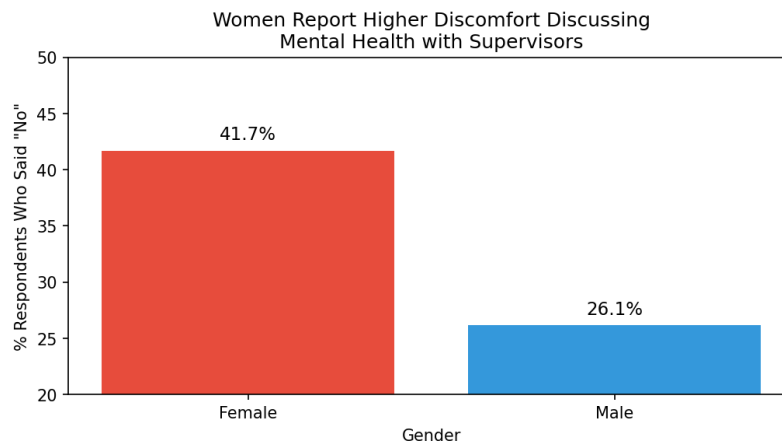


Figure 11—Bias disclosure

Bias hypothesis: As shown in the bar chart, women are much less comfortable discussing mental health with supervisors than men, suggesting gender bias.

Manipulations: I used a bar chart and plotted only the “No” responses (uncomfortable). I set the y-axis to start at 20 instead of 0, which makes the gap between female (41.7%) and male (26.1%) look much larger. I also added percentage labels on each bar and used a title that highlights women’s higher discomfort.

5 AVERAGES TABLE

	Mean	Median	Mode
Original dataset	0.70 (Male)	1 (Male)	1 (Male)
Reduced dataset (50%)	0.68 (Male)	1 (Male)	1 (Male)
Difference	Difference	No difference	No difference

Table 10—Comparison Between Original and Reduced Datasets

6 REDUCED DATASET

6.1 Table and Histogram

Gender	Maybe	No	Yes
Female	9	22	15
Male	28	33	35

Table 11—Reduced Disclosure Gender Analysis Data

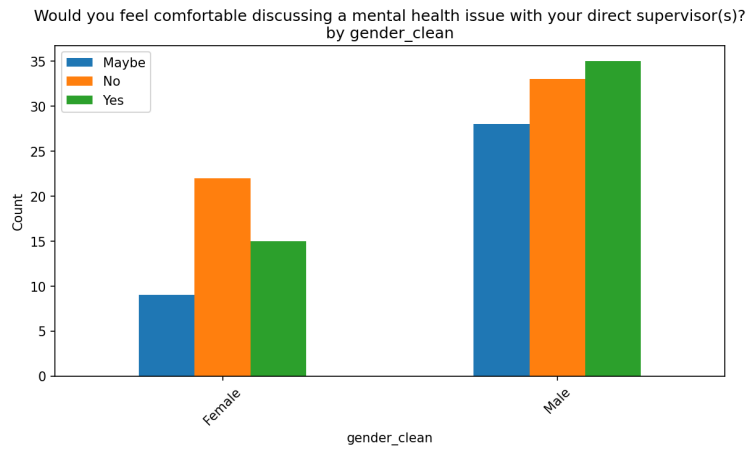


Figure 12—Bias disclosure

6.2 Difference

	Maybe	No	Yes	Total
Full-Female	24	35	25	84
Full-Male	64	52	83	199
Reduced-Female	9	22	15	46
Reduced-Male	28	33	35	96

Table 12—Distribution Comparison Across Full and Reduced Gender Datasets

The reduced dataset has 142 people (46 female, 96 male), about half of the original 283 (84 female, 199 male). This is expected from a 50% random sample.

Men still make up most of the sample (68% vs 70% in the full data). The overall pattern did not change: women are still more likely to say “No” and less likely to say “Yes” than men.

Some numbers shifted because the sample is smaller. For example, 47.8% of women said “No” in the reduced data vs 41.7% in the full data. Male “No” also went up (26.1% to 34.4%) but these are small sample changes, not a new trend.

6.3 Harmed members

Women are already the smaller group in this survey. Random sampling kept that imbalance, so male responses still dominate the reduced dataset. Women report more discomfort discussing mental health with supervisors than men in both datasets. If decisions were based only on the reduced sample, women’s experiences could get less attention because there are fewer female responses overall. For that reason, women are more likely to be harmed by using this random sample.

REFERENCES

- [1] Civil Rights Act of 1964, Title VII, 42 U.S.C. § 2000e et seq. (1964). URL: <https://www.eeoc.gov/statutes/title-vii-civil-rights-act-1964>.
- [2] Age Discrimination in Employment Act of 1967, 29 U.S.C. § 621 et seq. (1967). URL: <https://www.eeoc.gov/statutes/age-discrimination-employment-act-1967>.
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- [4] Stony Brook University Libraries. (n.d.). “Data cleaning and wrangling: Cleaning”. Research Guides. URL: <https://guides.library.stonybrook.edu/data-cleaning-and-wrangling/cleaning>.
- [5] The Molecular Sciences Software Institute. (n.d.). “Data processing & cleaning: Pandas data cleaning”. Python Data Science for Chemistry. URL: https://education.molssi.org/python-data-science-chemistry/data_processing_cleaning/pandas-datacleaning.html.